RETAIL ANALYTICS PORTFOLIO PROJECT

Tankhoone Fashion Retail — Tehran 2023

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Demand Analysis	Inventory Planning	Financial Metrics	Scenario Modeling	Dashboard Design

PROJECT OVERVIEW

This project is a self-directed retail analytics case study designed to demonstrate the analytical and operational skills required for retail analytics and supply chain planning roles in Canadian retail. Using a synthetic Tehran 2023 fashion retail dataset of 40 SKUs and approximately 1,215 transactions, I designed and executed a complete end-to-end analysis across eight structured chapters — from raw data cleaning through advanced scenario modeling and executive dashboards.

The project was built entirely in Microsoft Excel without macros or Power Query, deliberately chosen to demonstrate formula-level analytical thinking rather than tool dependency. Every formula, assumption, and finding is documented with clear business reasoning — the same standard expected in a professional retail environment.

PROJECT SCOPE

Dataset	Tool	SKUs Analyzed	Findings Documented
Tehran 2023 Fashion Retail	Microsoft Excel	40 SKUs | 8 Categories	26 Documented Findings

KEY FINDINGS

The analysis surfaced 26 documented findings across demand, inventory, financial performance, and advanced scenario modeling. The five most strategically significant are:

1

Revenue Is Dangerously Concentrated — and Capital Is Misallocated

Co-ord Sets alone account for 33% of total revenue (593M IRR). The top 20 SKUs by revenue drive 70% of total sales — a classic concentration risk. However, when the same assortment is evaluated by GMROI (capital efficiency), only 5 SKUs qualify as Class A versus 20 by revenue. Twenty-two SKUs are Class C by GMROI — meaning the business is tying up significant buying capital in products that generate poor returns on investment.

2

The Store Is Already in a Replenishment Emergency

Average Days of Supply across the 40-SKU assortment is 12 days. Lead time is 14 days. The store cannot replenish stock fast enough to cover current demand at normal rates — before a single spike scenario is considered. Six SKUs have already fallen below their Reorder Point and require immediate action: Beige Co-ord Set, Red Sweatshirt, Cable Knit Vest White, and three others.

3

Demand Is Volatile Across the Entire Assortment — No Category Is Stable

Every product category shows a High Coefficient of Variation (CV), meaning demand is unpredictable across the full assortment — not just individual SKUs. This is the structural challenge of Tehran fashion retail: no category provides a stable, forecastable demand pattern. Traditional reorder triggers based on average demand are insufficient. Safety stock buffers must account for high variability at all times.

4

A 25% Demand Spike Immediately Threatens 6 SKUs — the Same 6 Already Below ROP

Sensitivity modeling shows that even the most conservative demand spike scenario (25% above normal) puts 6 SKUs in immediate Danger — meaning current stock falls below the spike-adjusted Reorder Point. Critically, these are the same 6 SKUs already flagged for reorder under normal demand. These SKUs sit at the intersection of low stock, high demand sensitivity, and high revenue concentration — they represent the highest single risk in the assortment.

5

35% of the Assortment Is Dead Weight — Consuming Capital With No Return

A Margin-Velocity matrix classifying all 40 SKUs across gross margin and sales velocity reveals 14 Dead Weight SKUs (low margin, low velocity) and 9 Sleepers (good margin, not selling). Combined, 57.5% of the assortment requires either immediate markdown action or investigation. Meanwhile 11 Star SKUs with high margin and high velocity are likely under-protected given the current inventory shortfalls.

STRATEGIC RECOMMENDATIONS

Based on the full analysis, three immediate actions are recommended for the business:

1

Trigger Emergency Reorder for the 6 Critical SKUs Today

PRD-001, PRD-003, PRD-009, PRD-015, PRD-030, and PRD-038 must be reordered immediately. These SKUs are below ROP under normal demand, already in Danger under a 25% spike, and include top-revenue Co-ord Sets and Sweatshirts. With a 14-day lead time and only 2–5 units on hand per SKU, any further delay risks stockout during the next selling cycle. Buyer should escalate these to supplier within 24 hours.

2

Redirect Buying Capital From Dead Weight to Stars

Fourteen SKUs deliver neither sales velocity nor margin. The capital tied up in these SKUs — inventory investment plus shelf space — should be redirected toward the 11 Star SKUs that deliver on both dimensions. Before the next buying cycle, conduct a markdown review

of the 14 Dead Weight SKUs and investigate the 9 Sleepers to determine whether poor performance reflects pricing, placement, or genuine demand weakness. GMROI-based buying decisions should supplement revenue-based ABC going forward.

3

Implement a Minimum Data Standard for Inventory Tracking

Four SKUs (PRD-008, PRD-019, PRD-031, PRD-033) have no Units_On_Hand recorded in the system. Stock status cannot be assessed for these SKUs — they are analytically invisible. Additionally, order dates are not captured at the transaction level, preventing accurate lead time measurement. At minimum, the business needs to capture Units_On_Hand and order date at the point of receipt. Without this data, inventory planning decisions are based on assumptions rather than facts.

ANALYTICAL APPROACH & METHODS

Chapter	Method & Output
Ch1–2	Data cleaning and validation — ghost blank resolution, date serial correction, referential integrity checks across three source tables
Ch3	Exploratory Data Analysis — 4 pivot tables, 4 charts, 9 findings on seasonality, channel mix, and category concentration
Ch4	Demand Analysis — Coefficient of Variation per SKU, trend analysis via SLOPE function, moving averages
Ch5	Inventory Planning — Safety Stock (Z=1.65, 95% service level), Reorder Point, Days of Supply, Stock Status classification
Ch6	Financial Metrics — Sell-Through Rate, Gross Margin %, Revenue-Based ABC (Pareto), GMROI classification
Ch7	Advanced Analysis — Demand spike sensitivity (25/50/75%), one-variable Data Table for lead time modeling, stockout timeline modeling, Unit Margin-Velocity 2x2 matrix
Ch8	Dashboard Design — Business Performance dashboard for senior leadership; Operations & Inventory dashboard for buying managers and supply chain planners

EXCEL SKILLS DEMONSTRATED

Formula & Feature Skills	Analytical & Design Skills
• XLOOKUP, SUMIF, COUNTIF, IFERROR • LET() for intermediate calculation storage • Nested IF with AND() for multi-condition logic • RANK.EQ with cumulative SUMIF for ABC	• Demand volatility analysis (CV methodology) • Safety stock and ROP calculation • ABC classification (revenue and GMROI) • Sensitivity and scenario analysis

• MEDIAN, AVERAGE, COUNTA, ROUND, SQRT • Mixed and absolute cell references ($) • Pivot Tables and PivotCharts • One-variable Data Table (What-If Analysis) • Conditional Formatting — multi-rule • Paste as Picture for dashboard locking	• Unit Margin-Velocity 2x2 matrix • Lead time impact modeling (Data Table) • Stockout timeline and cost quantification • Two-audience dashboard design • Cross-chapter insight synthesis • Assumption documentation and business framing

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This document accompanies the Tankhoone Fashion Retail Analytics Excel workbook.

The workbook contains all source data, calculated sheets, advanced analysis, and two executive dashboards referenced in this narrative.